

GPU Computing

Exercise 2

Luis Wenzel
Miro Joensuu
Daniel Bayer
Noah Schneider

1 Global Memory - cudaMemcpy

2 Global Memory - cudaMemcpy

2.1 cudaMemcpy Task

In this Task, the use of `cudaMalloc(&hmem, dat);` and `cudaMemcpy(hmem, dmem, dat, cudaMemcpyDeviceToDevice);` was analyzed. That way we copy data from device to device. In the Graph 1 we can see, that a saturation of the copying speed occurs on higher data transportation rates. In our Graph, we also show the other copying speed of the other data. If you would like to take a look at our data, the data of the graphs is in folder (`exc03/Latex_File/data/analysis`).

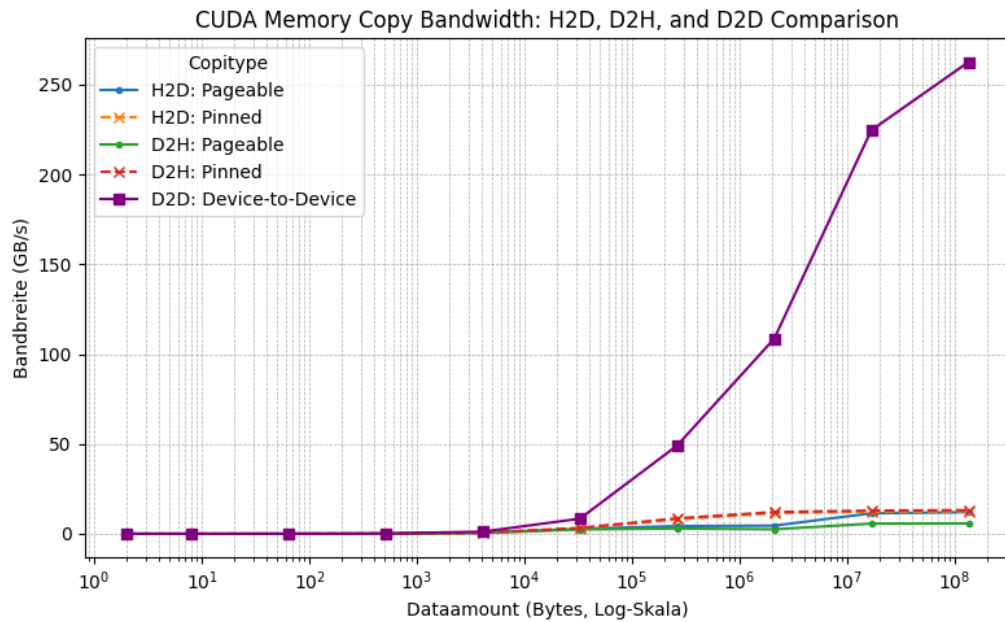


Figure 1: Speed difference of Device to Device memory copy

3 Global Memory - coalesced thread access

3.1 Break-even Kernel Startup Time

4 Global Memory -thread access with stride

4.1 Global Memory -thread access with stride

5 Global Memory -thread access with offset

5.1 Global Memory -thread access with offset

6 Presenting

Willing to present: